SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
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Date:	8/10/2026	**Duration:**	60 minutes

Code of Conduct

I __T. Deekshith (192425145) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __T. Deekshith _____

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format:

As a <user>, I want <feature>, so that <benefit>.

3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.

4. **Prioritize the Product Backlog**

- Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.

5. **Plan the Sprint Backlog**

- Select high-priority user stories for the first sprint and divide them into actionable development tasks.

6. **Estimate Effort Using Story Points**

- Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.

7. **Present the Sprint Goal and Expected Deliverables**

- Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Introduction

Bovine mastitis is an inflammation of the udder in dairy cattle that affects milk production, milk quality, and animal health. In Indian dairy farms, delayed detection can increase treatment costs and cause economic losses to farmers.

The proposed AI-based system uses animal health records, milk-yield changes, previous mastitis history, and other available indicators to predict the risk of mastitis at an early stage. It generates risk alerts to help farmers identify animals that may need veterinary attention. Agile Scrum is used to develop the system in small increments, allowing regular testing, stakeholder feedback, and continuous improvement.

1. Project Vision and Stakeholders

1.1 Project Vision

The vision is to develop an AI-based system that predicts the early risk of bovine mastitis in Indian dairy farms. It aims to support early observation, reduce avoidable production losses, improve animal welfare, and help farmers make informed decisions with veterinary guidance.

1.2 Project Objectives

- Collect and maintain digital health and milk-production records for dairy cattle.
- Develop a predictive model using suitable historical and current animal data.
- Identify cows that may be at increased risk of mastitis.
- Generate timely alerts for animals requiring closer observation.
- Evaluate model performance using appropriate machine-learning metrics.
- Protect farm records and provide a simple, user-friendly interface.

1.3 Stakeholders

- Dairy Farmers: Enter animal information and respond to risk alerts.
- Veterinarians: Review alerts and provide professional advice.
- Dairy Cooperatives: Support farms and coordinate animal-health information.
- Data Scientists: Prepare datasets, train models, and evaluate predictions.
- Software Developers: Build the application, database, and prediction services.
- Testing Team: Verify functionality, accuracy-related requirements, and security.
- Product Owner: Prioritize requirements and approve completed features.
- System Administrator: Manage user access and protect stored records.

1.4 Roles and Responsibilities

Stakeholder	Responsibility
Farmers	Record animal health and milk-production details.
Veterinarians	Review predictions and confirm health concerns.
Data Scientists	Develop and evaluate the predictive model.
Developers	Implement and integrate system features.
Testers	Test the system and report defects.
Product Owner	Manage priorities and project requirements.

1.5 Scrum Team

The Scrum team consists of a Product Owner, Scrum Master, software developers, a machine-learning engineer, and a tester. The team follows two-week sprints, delivering and reviewing small working features at the end of each sprint.

2. Create Epics and User Stories

Epics represent major system functions, while user stories describe specific features from the user's perspective.

Epic 1 – Animal Registration and Records

- US-1: As a farmer, I want to register each cow with a unique ID so that its health records can be maintained.
- US-2: As a farmer, I want to enter milk yield and health observations so that changes can be monitored.
- US-3: As a veterinarian, I want to view previous mastitis cases so that I can understand the animal's health history.

Epic 2 – Data Preparation

- US-4: As a data scientist, I want to validate animal records so that incorrect data does not affect predictions.
- US-5: As a data scientist, I want to identify missing and duplicate records so that data quality can be improved.
- US-6: As a researcher, I want clear data labels so that model training uses consistent information.

Epic 3 – Predictive Model Development

- US-7: As a machine-learning engineer, I want to train a baseline model so that mastitis risk can be estimated.
- US-8: As a researcher, I want to evaluate model performance so that prediction errors can be identified.
- US-9: As a veterinarian, I want to view important risk indicators so that predictions can be interpreted.

Epic 4 – Risk Prediction and Alerts

- US-10: As a farmer, I want to check a cow's mastitis risk so that I can take timely action.
- US-11: As a farmer, I want alerts for high-risk animals so that they can receive closer attention.
- US-12: As a veterinarian, I want to record alert outcomes so that predictions can be reviewed later.

Epic 5 – Dashboard and Reporting

- US-13: As a farm owner, I want to view animals by risk category so that I can monitor herd health.
- US-14: As a cooperative supervisor, I want summary reports so that I can review alerts and outcomes.

Epic 6 – Security and Reliability

- US-15: As an administrator, I want role-based access so that only authorised users can view records.
- US-16: As a farmer, I want to know when prediction services are unavailable so that I do not rely on outdated results.
- US-17: As a project owner, I want prediction timestamps and model versions recorded so that results can be traced.

Summary of Epics

Epic	Purpose
Animal Records	Maintain cow details and health history.
Data Preparation	Improve data quality and consistency.
Predictive Modelling	Train and evaluate the AI model.
Risk Alerts	Identify animals requiring attention.
Dashboard	Display risk information and reports.
Security	Protect records and track predictions.

3. Define Acceptance Criteria

Acceptance criteria define the conditions that must be satisfied before a user story is accepted. The Given–When–Then format makes requirements clear and testable.

US-1: Register a Cow

- Given an authorised user enters valid animal details, When the form is submitted, Then the system saves the record with a unique ID.
- Given the animal ID already exists, When the user submits it again, Then the system displays a duplicate-record warning.

US-2: Record Animal Observations

- Given a registered cow exists, When the user enters milk-yield and health details, Then the system saves the observation with its date.
- Given required information is missing, When the form is submitted, Then the system identifies the missing fields.

US-4: Validate Data

- Given a record contains invalid values, When validation runs, Then the system flags the incorrect information.
- Given a dataset is checked, When validation finishes, Then the system reports accepted and rejected records.

US-5: Identify Missing Data

- Given a dataset is available, When quality analysis runs, Then missing values and possible duplicates are reported.

US-6: Define Data Labels

- Given mastitis outcome labels are required, When the data guidelines are reviewed, Then the accepted evidence and labeling rules are documented.

US-7: Train a Baseline Model

- Given an approved labelled dataset is available, When model training runs, Then the system produces a model and records its evaluation results.
- Given the data is insufficient, When training is attempted, Then the limitation is reported.

US-8: Evaluate Model Performance

- Given a trained model is available, When evaluation runs, Then metrics such as recall, precision, F1-score, and specificity are reported where applicable.

US-10: Generate Risk Predictions

• Given sufficient recent animal data is available, When a prediction is requested, Then the system displays a risk category and prediction timestamp.
• Given required data is missing, When prediction is requested, Then the system explains why a reliable prediction cannot be generated.

US-11: Generate Alerts

• Given a prediction meets the approved high-risk threshold, When the result is saved, Then the system generates an alert for review.
• Given an alert is displayed, When the farmer opens it, Then the system provides an option to record the follow-up status.

US-15: Secure User Access

- Given a user is not authorised, When the user requests protected animal records, Then access is denied.

Definition of Done

A feature is considered complete when its acceptance criteria pass, relevant tests succeed, code is reviewed, documentation is updated, and the Product Owner accepts the feature. Model-related work must also document its dataset, evaluation results, and limitations.

4. Prioritize the Product Backlog

The product backlog contains all features required for the project. Items are prioritised according to their importance, user value, dependencies, and risk.

Priority	Backlog Item	Business Value	Reason
High	User authentication	High	Protects farm records.
High	Animal registration	High	Establishes unique cow identities.
High	Health data entry	High	Provides input for predictions.
High	Data validation	High	Prevents incorrect records.
High	Missing-data analysis	High	Identifies data-quality issues.
High	Data labeling guidelines	High	Supports reliable model training.

High	Baseline AI model	High	Provides initial risk predictions.
High	Model evaluation	High	Measures prediction performance.
High	Risk prediction and alerts	High	Helps farmers identify animals requiring attention.
Medium	Animal history	Medium	Supports health assessment.
Medium	Dashboard and reports	Medium	Improves herd monitoring.
Medium	Alert outcome recording	Medium	Supports future evaluation.
Low	Automatic sensor integration	Medium	Reduces manual data entry in future versions.
Low	Multilingual interface	Medium	Improves accessibility for more farmers.

4.1 Prioritisation Justification

High-priority features are developed first because they form the foundation of the system. Animal records, data validation, and reliable labels are necessary before building a trustworthy predictive model. Model evaluation is completed before operational alerts are relied upon. Dashboards, sensor integration, and additional language support can be developed in later sprints.

5. Plan the Sprint Backlog

The sprint backlog contains the selected user stories and the development tasks planned for the current sprint.

5.1 Sprint 1 Overview

Sprint 1 focuses on developing the basic data-management system. The team will implement animal registration, health-record entry, data validation, and secure access. It will also prepare initial data-quality reports and documentation for future AI model development.

The sprint is planned for two weeks. The following hours are illustrative estimates for a small development team.

User Story	Development Tasks	Estimated Hours
US-15: User authentication	Create login, define roles, and test access permissions.	18
US-1: Animal registration	Design database, create registration form, and validate unique IDs.	20
US-2: Health records	Create observation forms, store records, and display animal history.	22
US-4: Data validation	Define validation rules and test invalid values.	18
US-5: Data-quality report	Analyse missing values and identify duplicate records.	14
US-6: Data documentation	Define data fields and prepare mastitis labeling guidelines.	16
Project setup	Configure the repository, testing process, and integration.	14
Integration and testing	Test features, fix defects, and prepare documentation.	10
Total		132 hours

5.2 Sprint Backlog Justification

These tasks are selected because they establish the data foundation needed for machine learning. Animal registration identifies each cow, health records capture important observations, and validation improves data reliability. Security protects stored information, while data-quality analysis identifies issues that could affect future model performance. The remaining sprint capacity is reserved for unexpected issues and integration work.

5.3 Sprint Task Workflow

1. Day 1: Finalise requirements and set up the project.
2. Days 2–3: Develop authentication and animal registration.
3. Days 4–5: Implement health-record entry and validation.
4. Days 6–7: Generate data-quality reports and document labels.
5. Day 8: Integrate features, test the system, and prepare the demonstration.

6. Estimate Effort Using Story Points

Story points estimate the relative effort, complexity, and uncertainty of each user story. The team uses Planning Poker and the Fibonacci sequence: 1, 2, 3, 5, 8, and 13.

User Story	Story Points	Justification
US-1: Animal registration	5	Requires database design, form development, and validation.
US-2: Health-record entry	8	Includes observation storage, timestamps, and history display.
US-3: Animal history	3	Mainly involves retrieving and displaying records.
US-4: Data validation	5	Requires validation rules and testing.
US-5: Data-quality analysis	3	Involves missing-value analysis and duplicate reporting.
US-6: Data labeling	5	Requires documentation and veterinary consultation.
US-7: Baseline AI model	8	Requires data preparation, training, and evaluation.
US-8: Model evaluation	5	Requires performance metrics and error analysis.
US-9: Risk explanations	8	Requires interpretation of important model indicators.
US-10: Risk prediction	8	Integrates the model with the application.
US-11: Alerts	5	Requires thresholds, alert generation, and status tracking.
US-12: Outcome recording	3	Stores follow-up information.
US-13: Dashboard	5	Displays risk categories and alert history.
US-14: Reports	5	Requires data aggregation and reporting.
US-15: Authentication	5	Requires login, permissions, and security testing.
US-16/17: Reliability and traceability	3	Records service status, prediction time, and model version.

6.1 Estimation Justification

Simple features receive fewer points, while complex features involving data processing and model integration receive more points. For example, animal registration is estimated at five points, whereas health-record entry receives eight points because it involves several connected operations.

The Sprint 1 stories selected in Section 5 total approximately 31 story points. These estimates are provisional and should be adjusted according to actual team capacity and progress.

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

The goal of Sprint 1 is to develop a secure and reliable data-management foundation for the AI-based bovine mastitis forecasting system. By the end of the sprint, users should be able to register cows, enter health observations, validate records, and view animal history. The team will also prepare data-quality reports and labeling guidelines for future predictive modelling.

7.2 Expected Sprint Deliverables

Deliverable	Description
Animal Registration Module	Registers cows using unique identification numbers.
Health-Record Module	Stores dated milk-yield and health observations.
Authentication Module	Restricts access to authorised users.
Data Validation Module	Identifies missing and invalid information.
Data-Quality Report	Summarises missing values and possible duplicates.
Data Dictionary	Documents the meaning and units of data fields.
Mastitis Labeling Guidelines	Defines how mastitis outcomes will be identified in the dataset.
Testing Report	Documents test cases and results.
Sprint Demonstration	Presents completed features to stakeholders.

7.3 Sprint Review

At the end of the sprint, the team demonstrates the completed features to the Product Owner, farmers, and veterinary representatives. Their feedback is collected to identify missing requirements, usability issues, and improvements. New requirements are added to the product backlog.

7.4 Sprint Retrospective

The team reviews what went well, identifies problems encountered during development, and discusses improvements. Possible improvements include clearer requirements, better testing, earlier veterinary consultation, and more accurate task estimation. These improvements are applied during the next sprint.

8. Future Sprint Roadmap

Sprint	Main Activities	Expected Outcome
Sprint 1	Animal records, security, and data validation.	Working data-management foundation.
Sprint 2	Data preparation and baseline model training.	Initial model and evaluation report.
Sprint 3	Risk prediction API and alert generation.	Integrated prediction workflow.
Sprint 4	Dashboard and alert review features.	Usable farmer-facing prototype.
Sprint 5	Farm-level validation and performance testing.	Evaluation of model readiness and limitations.
Future Sprints	Sensor integration and multilingual support.	Improved accessibility and automation.

The roadmap may change according to data availability, stakeholder feedback, and model performance. The system should not be treated as a validated forecasting solution until appropriate evaluation has been completed.

9. Agile Workflow

The project follows an iterative Agile Scrum process:

1. Define the project vision.
2. Identify stakeholders and requirements.
3. Create epics and user stories.
4. Define acceptance criteria.
5. Prioritise the product backlog.
6. Estimate story points.
7. Select tasks during sprint planning.

8. Develop and test the software.
9. Demonstrate the completed increment.
10. Conduct the sprint retrospective.
11. Update the backlog and plan the next sprint.

This workflow supports continuous improvement and helps the team respond to changing project requirements.

10. Risks, Assumptions, and Mitigation

Risk	Impact	Mitigation
Missing animal records	Reduces data quality.	Validate data and report missing values.
Incorrect mastitis labels	Produces unreliable model evaluation.	Consult veterinary experts and document labeling rules.
Model prediction errors	May cause missed or unnecessary alerts.	Evaluate sensitivity, precision, and other relevant metrics.
Differences between farms	Model performance may vary across settings.	Validate using data from different farms where available.
Data privacy issues	May expose farm information.	Use authentication and role-based access.
Limited internet access	Interrupts system usage.	Consider lightweight and offline-tolerant features.
Model performance changes	Predictions may become less reliable over time.	Monitor outcomes and reassess model performance.
Misinterpretation of alerts	Users may treat predictions as diagnoses.	Clearly explain limitations and recommend veterinary review.

11. Success Measures

The project's success will be evaluated using the following measures:

- Completeness and accuracy of animal records.
- Percentage of records passing data validation.
- Successful completion of registration and observation-entry tasks.
- Predictive performance measured using recall, precision, F1-score, and specificity.

- Reliability and timeliness of risk alerts.
- Usability feedback from farmers and veterinarians.
- Security and access-control test results.
- Performance validation across farms where suitable data is available.

Actual model performance can only be reported after training and testing on an appropriate dataset.

12. Conclusion

The AI-Based Predictive Modelling for Early Forecasting of Bovine Mastitis in Indian Dairy Farms project aims to support early identification of cows that may be at risk of mastitis. Agile Scrum divides development into manageable sprints, beginning with animal records, data validation, and security before progressing to predictive modelling and alerts.

User stories, acceptance criteria, backlog prioritisation, and story-point estimation help organise development and measure progress. Future sprints will focus on model training, performance evaluation, risk alerts, and dashboards. By combining reliable data, responsible AI evaluation, and veterinary guidance, the proposed system can support better animal-health monitoring and more informed dairy-farm management.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25